

02_Requirement_Analysis

□ 1. Introduction

Requirement Analysis is the most important phase of any automation project. It defines **what needs to be tested, why it is tested, and how it will be validated.**

For the OrangeHRM automation project, the requirements are derived from the core functionalities of the application such as login, password recovery, and navigation modules.

□ 2. Functional Requirements

Functional requirements describe the **behavior of the application** that must be validated through automation.

◆ 2.1 Login Module Requirements

The system must allow users to log in with valid credentials.

✓ Requirements:

- User should be able to login using valid username and password
- System should display dashboard after successful login
- System should show error message for invalid credentials
- System should validate empty username field
- System should validate empty password field

✓ Expected Behavior:

- Valid login → Redirect to Dashboard page
 - Invalid login → Show error message “Invalid credentials”
 - Empty fields → Show validation messages
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◆ 2.2 Forgot Password Module Requirements

The system must allow users to reset their password using registered username.

✓ Requirements:

- User should be able to navigate to Forgot Password page
- System should accept valid username and send reset instructions
- System should display warning for invalid username
- System should show error for empty username
- User should be able to cancel and return to login page

✓ **Expected Behavior:**

- Valid username → Success message displayed
 - Invalid username → Contact HR warning message
 - Empty input → Missing username validation message
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◆ 2.3 Navigation Module Requirements

The system should allow navigation after successful login.

✓ **Requirements:**

- User should be able to access dashboard after login
- User should be able to navigate to Apply Leave page
- Menu navigation should work using hover actions
- Page transitions should be smooth without errors

✓ **Expected Behavior:**

- Dashboard → Apply Leave navigation should open correct page
 - Page header should match expected text “Apply Leave”
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□ 3. Non-Functional Requirements

Non-functional requirements define **quality aspects** of the system.

◆ 3.1 Performance

- Application should load within acceptable time
 - Page navigation should not be delayed
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◆ 3.2 Usability

- UI should be user-friendly

- Error messages should be clearly visible
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◆ 3.3 Compatibility

- Application should work on Chrome and Firefox browsers
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◆ 3.4 Maintainability (Automation Framework)

- Framework should be modular using Page Object Model
 - Code should be reusable and easy to maintain
 - Test cases should be independent
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□ 4. Automation Requirements

These define what must be automated in the framework.

✓ Must Automate:

- Login functionality (valid + invalid cases)
 - Forgot password workflow
 - Navigation to Apply Leave page
 - UI validation messages
 - Page header validations
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✗ Not Automated:

- Backend database validation
 - Performance/load testing
 - API level testing
 - Security penetration testing
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□ 5. Test Data Requirements

◆ Login Test Data:

- Username: Admin
- Password: admin123

- Invalid usernames/passwords for negative testing
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◆ **Forgot Password Test Data:**

- Valid Username: Admin
 - Invalid Username: random strings
 - Empty input scenarios
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□ **6. Environment Requirements**

◆ **Hardware:**

- Standard laptop/desktop system

◆ **Software:**

- Windows OS
 - Eclipse IDE
 - Java JDK
 - Selenium WebDriver
 - TestNG Framework
 - ChromeDriver / GeckoDriver
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□ **7. Dependencies**

- Internet connection for accessing OrangeHRM demo site
 - Browser drivers installed correctly
 - Stable Selenium WebDriver setup
 - TestNG configured in Eclipse
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□ **8. Risk Analysis**

⚠ **Possible Risks:**

- UI changes in OrangeHRM site may break locators
- Browser compatibility issues
- Test failures due to synchronization issues
- Network downtime during execution

✓ Mitigation:

- Use stable locators (ID, CSS)
- Implement proper waits (future enhancement)
- Maintain updated driver versions

□ 9. Summary

Requirement Analysis clearly defines:

- What modules are tested (Login, Forgot Password, Navigation)
 - What scenarios are validated (positive & negative cases)
 - What is included and excluded from automation scope
 - What environment and tools are required
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